

Octacosanol Supplementation Increases Running Endurance Time and Improves Biochemical Parameters After Exhaustion in Trained Rats

Hyunjung Kim,¹ Sunghee Park,¹ Dae Suk Han,² and Taesun Park¹

¹*Department of Food and Nutrition, Yonsei University, Seoul; and* ²*Korea Food Research Institute, Kyunggi-do, Korea*

ABSTRACT This study evaluated the effects of octacosanol on running performance and related biochemical parameters in exercise-trained rats run to exhaustion on a treadmill. Male Sprague–Dawley rats were randomly assigned to one of three groups—sedentary control group (SC), exercise-trained control group (EC), and exercise-trained, octacosanol-supplemented group (EO)—and raised on either control or octacosanol (0.75%)-supplemented diet with (or without for SC rats) exercise-training for 4 weeks. EC rats ran 184% longer until exhaustion than SC rats ($P < .01$), while octacosanol-supplemented trained rats ran 46% longer than EC rats ($P < .05$). Under the exhausted state immediately following the running performance test, EO rats exhibited significantly higher plasma ammonia and lactate concentrations compared with the values for EC rats ($P < .05$). Although EO rats ran significantly longer until exhausted, their plasma glucose level and gastrocnemius muscle glycogen concentration were not significantly different from those of EC rats. Dietary supplementation of octacosanol resulted in significantly higher creatine phosphokinase activity in plasma (44% increase, $P < .01$) and citrate synthase activity in muscle (16% increase, $P < .01$) of exercise-trained rats. These results suggest that the ergogenic properties of octacosanol include the sparing of muscle glycogen stores and increases in the oxidative capacity in the muscle of exercise-trained rats.

KEY WORDS: • *exercise-trained rats* • *glycogen* • *lactate* • *octacosanol* • *running performance*

INTRODUCTION

OVER THE PAST 10 YEARS, nutritional supplements have become increasingly popular among the general public worldwide, and many supplements are taken as ergogenic aids, a term used for substances that enhance athletic performance and increase stamina and exercise capacity. Ergogenic aids are believed to increase athletic performance by either renewing or increasing energy stores in the body, modulating the biochemical reactions contributing to fatigue, or maintaining optimal body weight.¹

Octacosanol [$\text{HO-CH}_2\text{-(CH}_2\text{)}_{26}\text{CH}_3$], a high-molecular-weight, primary aliphatic alcohol, is the main component of a natural wax product that naturally occurs in fruits and leaves and on the surface of plants and whole seeds. Most studies evaluating the efficacy of octacosanol have used either a wheat-germ oil extract (policosanols) or a natural mixture of primary alcohols isolated from sugar cane wax (*Saccharum officinarum* L.), of which octacosanol is the main component.^{2,3} Recent studies in humans and rabbits with hypercholesterolemia^{4–9} have demonstrated that octacosanol is involved in the modulation of cholesterol metabolism, lowering blood low-density lipoprotein-cholesterol and elevat-

ing high-density lipoprotein-cholesterol concentrations. In addition to the cholesterol-lowering effects, anti-aggregatory properties and the cytoprotective activity of octacosanol have been investigated.

Ergogenic properties of octacosanol were first postulated by Cureton¹⁰ from the observation that extracts from wheat-germ oil (octacosanol) had beneficial effects on the physical performance of athletes. Another early study on octacosanol involved the investigation of its effects on metabolic responses to submaximal cycle ergometry, grip and chest strength, and reaction time in human subjects.² More recently, studies^{11,12} have suggested beneficial effects of octacosanol on physical performance in different systems. Most of these studies assessing the use of octacosanol on athletic performance have been conducted in humans using experimental designs lacking randomization, and appear to be nonspecific in terms of sports regimens and diet. Also, many claims for improving athletic performance as well as its underlying mechanism have yet to be proven in an appropriate and controlled animal model. The present study was, therefore, undertaken to evaluate effects of octacosanol, used as a dietary supplement, on endurance exercise performance in exercise-trained rats. Several biochemical parameters, such as blood concentrations of lactate, ammonia, glucose, and inorganic phosphorus, along with muscle glycogen concentration and enzyme activities involved in the energy production pathway during exercise have been assessed.

Received 12 May 2003. Accepted 26 June 2003.

Address reprint requests to: Taesun Park, Ph.D., Department of Food and Nutrition, Yonsei University, 134 Shinchon-dong, Sudaemun-ku, Seoul 120-749, South Korea, E-mail: tspark@yonsei.ac.kr

TABLE 1. COMPOSITION OF EXPERIMENTAL DIETS

	Level (g/kg)	
	CD	OD
Cornstarch	367.486	359.986
Dextrinized cornstarch	132.000	132.000
Sucrose	100.000	100.000
Casein	200.000	200.000
Corn oil	100.000	100.000
Cellulose	50.000	50.000
Mineral mix (AIN-93G-MX) ^a	35.000	35.000
Vitamin mix (AIN-93G-VX) ^b	10.000	10.000
L-Cystine	3.000	3.000
Choline bitartrate	2.500	2.500
<i>tert</i> -Butylhydroquinone	0.014	0.014
Octacosanol ^c	—	7.500

CD, control diet; OD, octacosanol-supplemented diet.

^aPrepared by ICN Biomedicals, Inc. (Costa Mesa, CA).

The mineral mixture contained (in g/kg of mix): calcium carbonate, anhydrous, 357; potassium phosphate, monobasic, 196; potassium citrate, tri-potassium, monohydrate, 70.78; sodium chloride, 74; potassium sulfate, 46.6; magnesium oxide, 24; ferric citrate, 6.06; zinc carbonate, 1.65; manganous carbonate, 0.63; cupric carbonate, 0.3; potassium iodate, 0.01; sodium selenate, anhydrous, 0.01025; ammonium paramolybdate, 4-hydrate, 0.00795; sodium metasilicate, 9-hydrate, 1.45; chromium potassium sulfate, 12-hydrate, 0.275; lithium chloride, 0.0174; boric acid, 0.0815; sodium fluoride, 0.0635; nickel carbonate, 0.0318; ammonium vanadate, 0.0066; powdered sucrose, 221.026

^bPrepared by ICN Biomedicals, Inc. The vitamin mixture contained (in g/kg of mix): nicotinic acid, 3; Ca pantothenate, 1.6; pyridoxine-HCl, 0.7; thiamine-HCl, 0.6; riboflavin, 0.6; folic acid, 0.2; biotin, 0.02; vitamin B-12 (cyanocobalamin) (0.1% in mannitol), 2.5; vitamin E (500 IU/g), 15; vitamin A (500,000 IU/g), 0.8; vitamin D₃ (cholecalciferol) (400,000 IU/g), 0.25; vitamin K (phyllloquinone), 0.075; powdered sucrose, 974.655.

^cIsolated from rice wax, and supplied by Sungkyun Biotech Co., Kyunggi-do, South Korea.

MATERIALS AND METHODS

Animals and experimental diets

Twenty-four male Sprague-Dawley rats weighing between 180 and 200 g were purchased from Daehan Laboratory Animal Research Center Co. (Chungcheongbuk-do,

Korea). Rats were acclimated to individual stainless steel cages in a room with controlled temperature (20–23°C), humidity (50–60%), and lighting (12-hour light-dark cycle) and fed a commercial pelleted rat chow for 5 days prior to initiating the experimental diets. The rats were randomly assigned to one of the following experimental groups: sedentary control group (SC; *n* = 8), exercise-trained control group (EC; *n* = 8), and exercise-trained, octacosanol-supplemented group (EO; *n* = 8).

Rats in the SC and EC groups were fed the control diet formulated using the AIN-93G purified rodent diet, and animals in the EO group were fed the same diet supplemented with 0.75% octacosanol for 4 weeks. Compositions of the experimental diets are as listed in Table 1. Animals were given free access to food and drinking water for the entire feeding period. Food consumption was recorded daily, and body weight gain was measured every third day between 1200 and 1300 h. This study adhered to the *Guide for the Care and Use of Laboratory Animals* developed by the Institute of Laboratory Animal Resources of the National Research Council,¹³ and was approved by the Institutional Animal Care and Use Committee of Yonsei University, Seoul, South Korea.

Exercise training protocol and running performance test

Rats in EC and EO groups were exercise-trained for the entire experimental period on a treadmill, at room temperature, 5 days/week. After being familiarized with a motor-driven rodent treadmill (Daejong Instrument Co., Seoul) for 2 days, animals were forced to run on a treadmill at 15 m/minute, 0% slope for 20 minutes/day during the first week. The intensity and duration of the exercise training were progressively increased until at week 4 the animals were running at 25 m/minute, 0% slope for 60 minutes/day as shown in Table 2.

At the end of the experimental period, animals in the SC, EC, and EO groups were all food-deprived for 2 hours, and then tested for their endurance exercise performance on a treadmill. The speed began with 12 m/minute, and was increased every 3 minutes by 3 m/minute until reaching 30 m/minute. Rats were run until they could no longer maintain the pace of the treadmill. Once a rat could not keep pace with the treadmill for up to 10 seconds, it was removed from the treadmill and placed in a supine position. If the rat did not right itself, it was considered exhausted and immediately

TABLE 2. EXERCISE TRAINING PROTOCOL ON A TREADMILL

Week	Intensity (m/min)	Duration (min)	Sum of running distance (m/week)
1	15	20	1,500
2	20	30	3,000
3	25	40	5,000
4	25	60	7,500

TABLE 3. BODY WEIGHT GAIN, FOOD EFFICIENCY RATIO, AND SKELETAL MUSCLE WEIGHTS OF RATS

	Group		
	SC	EC	EO
Body weight gain (g/4 weeks)	125 ± 5.52	107 ± 7.11	100 ± 3.73
Food efficiency ratio	0.28 ± 0.01	0.26 ± 0.02	0.23 ± 0.01
Gastrocnemius muscle (g/100 g of body weight)	646 ± 10.5	675 ± 7.7 ^a	677 ± 10.3
Soleus muscle (g/100 g of body weight)	35.5 ± 1.0	37.8 ± 1.3	39.3 ± 2.0

SC, sedentary control group; EC, exercise-trained control group; EO, exercise-trained, octacosanol-supplemented group. The food efficiency ratio was defined as the body weight gain for the experimental period (g/day)/food intake for the experimental period (g/day). Data are mean ± SEM values of eight rats.

^aSignificantly different compared with the value for SC rats by the independent samples *t* test at *P* < .05.

anesthetized for surgery. If rats were not exhausted, exercise was continued until exhaustion occurred. To avoid bias, rat identification was concealed from the investigator assessing the exhaustion criteria in all rats.

Sample collection

Immediately after the end of the running performance test, exhausted animals were anesthetized with ethyl ether, and the abdomen was opened. Blood samples were collected from the abdominal vein into a heparinized syringe, and plasma was prepared by a low-speed centrifugation (2,000 *g* for 10 minutes). Gastrocnemius and soleus muscles were surgically removed from both left and right hind limbs, weighed after trimming away connective tissues, freeze-clamped in liquid nitrogen, and stored at −80°C until analyses.

Analytical methods

Activity of plasma creatine phosphokinase (CK) (Bio Clinical System Corp., Anyang, Kyunggi-do, South Korea) and plasma concentrations of ammonia (Wako Pure Chemical, Osaka, Japan), lactate (Sigma, St. Louis, MO), glucose (Bio Clinical System Corp.), and inorganic phosphorus (Bio Clinical System Corp.) were analyzed using commercially available kits.

Citrate synthase (CS) activity was measured in soleus muscles as an index of the training-induced increase in oxidative capacity. Samples of thawed muscle tissue were minced and homogenized in ice-cold 100 mM potassium phosphate buffer with 0.05% bovine serum albumin (1:20 wt/vol, pH 7.4). Homogenates were then centrifuged at 400 *g* for 10 minutes. The supernatant was decanted and assayed for CS activity at 37°C using the procedure described by Srere.¹⁴ Activity of lactate dehydrogenase (LDH) was determined according to the method of Pesce *et al.*¹⁵ Gastrocnemius muscles were homogenized in ice-cold 100 mM potassium phosphate buffer (pH 7.4) using a Polytron homogenizer. The assay was performed in a reaction mixture containing 100 mM KH₂PO₄ (pH 7.5), 3.6 mM nicotinamide

adenine dinucleotide, and 3.3 mM pyruvate. Protein concentration was determined using reagents from Bio-Rad Laboratories (Hercules, CA). Enzyme activities were expressed in μmol (or nmol) mg of wet muscle^{−1} min^{−1}.

Glycogen content of the gastrocnemius muscle was determined by the method of Chun and Yin¹⁶ with minor modifications. The gastrocnemius muscle (0.2 g) was added to 0.5 mL of 30% KOH, and the vials were kept at 100°C for 20 minutes. After addition of 1.5 mL of anhydrous ethanol, the vials were centrifuged at 4,000 *g* for 15 minutes, and the supernatants were discarded. Distilled water (0.5 mL) and 1 mL of 0.2% anthrone (0.2 g of anthrone in 100 mL of freshly prepared 98% H₂SO₄) were added, and the vials were placed in boiling water for 20 minutes. The optical density of the reaction mixture was determined by photometry at 620 nm.

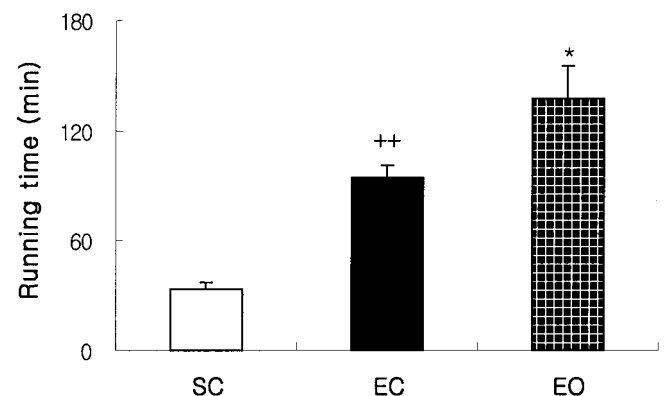


FIG. 1. Endurance exercise time to exhaustion in sedentary and trained rats: SC, sedentary control group; EC, exercise-trained control group; EO, exercise-trained, octacosanol-supplemented group. Data are mean ± SEM values of eight rats. ++Significantly different compared with the value for SC rats by the independent samples *t* test at *P* < .01. *Significantly different compared with the value for EC rats by the independent samples *t* test at *P* < .05.

TABLE 4. CORRELATION COEFFICIENTS BETWEEN RUNNING TIME UNTIL EXHAUSTED AND PLASMA CONCENTRATION OF GLUCOSE, AMMONIA, OR INORGANIC PHOSPHORUS, OR MUSCLE GLYCOGEN LEVEL OF RATS

Group	Glucose	Ammonia	Lactate	Inorganic phosphorus	Glycogen
SC	-0.214	0.565	0.822 ^a	0.653	-0.179
EC	-0.155	0.640	0.557	0.537	-0.811 ^a
EO	-0.699	0.330	0.576	0.196	-0.573

SC, sedentary control group; EC, exercise-trained control group; EO, exercise-trained, octacosanol-supplemented group. Data are mean values of eight rats.

^aSignificantly correlated by the Pearson correlation test at $P < .05$.

Statistical analyses

Data are expressed as mean $\pm$ SEM values of eight rats. All statistical analyses were performed using SAS/STAT version 8.1 (SAS Institute, Inc., Cary, NC). Significance of the effect of exercise training or dietary supplementation was analyzed by the independent samples t test at $P < .05$ or $P < .01$. Correlations between running time as an indicator of endurance exercise performance and plasma concentration of glucose, ammonia, lactate, or inorganic phosphorus, as well as muscle glycogen level, of rats were analyzed by the Pearson correlation test at $P < .05$.

RESULTS

The food efficiency ratio and skeletal muscle weights of rats are presented in Table 3. Exercise training on a treadmill for 4 weeks with or without dietary supplementation resulted in slightly lower values for cumulative body weight gain and food efficiency ratio compared with the SC rats, but these differences were not statistically significant ($P > .05$). Rats exercise-trained without dietary supplementation had a significantly higher gastrocnemius muscle mass ($P < .05$) with no difference in the soleus muscle weight compared with those for the SC animals ($P > .05$). Supplementing trained rats with octacosanol did not affect body weight gain, food efficiency ratio, or skeletal muscle weights compared with the EC animals.

In this study, both exercise training and octacosanol supplementation significantly affected running time to exhaustion (Fig. 1). In the SC rats, mean running time to exhaustion was 33.3 ± 3.8 minutes (range 26–59 minutes), whereas trained rats ran 184% longer, 94.5 ± 6.6 minutes (range 76–125 minutes), which was statistically significant ($P < .01$). The EO rats ran significantly ($P < .05$) longer (46%) than the trained rats without supplementation, with an average running time of 137.6 ± 17.5 minutes (range 90–248 minutes).

Correlations between running time, as an indicator of endurance exercise performance, and the plasma concentrations of diverse fatigue-related parameters as well as muscle glycogen concentration were analyzed in the SC, EC, and EO rats (Table 4). The positive correlation observed between plasma lactate concentration and running time to exhaustion was significant in the SC rats ($P < .05$), but was without statistical significance in the exercise-trained animals with or without dietary supplementation (EO and EC, $P > .05$). Running time to exhaustion was positively correlated, but not significantly ($P > .05$), with plasma concentrations of ammonia and inorganic phosphorus, and negatively correlated with plasma glucose concentrations (also not significant). There was an inverse correlation of running time with muscle glycogen concentration in all three groups, but it was only significant in the EC rats.

Table 5 shows the plasma concentrations of glucose, ammonia, lactate, and inorganic phosphorus in exhausted rats

TABLE 5. PLASMA CONCENTRATIONS OF AMMONIA, LACTATE, GLUCOSE, AND INORGANIC PHOSPHORUS IN EXHAUSTED RATS FOLLOWING THE RUNNING PERFORMANCE TEST

	Group		
	SC	EC	EO
Glucose (mg/dL)	221 $\pm$ 13.0	154 $\pm$ 8.83 ^a	141 $\pm$ 13.7
Ammonia (μ g/dL)	209 $\pm$ 11.2	193 $\pm$ 17.5	253 $\pm$ 12.1 ^b
Lactate (mg/dL)	26.6 $\pm$ 2.12	22.1 $\pm$ 1.49	27.2 $\pm$ 1.24 ^b
Inorganic phosphorus (mg/dL)	5.30 $\pm$ 0.14	5.80 $\pm$ 0.23	6.08 $\pm$ 0.28

SC, sedentary control group; EC, exercise-trained control group; EO, exercise-trained, octacosanol-supplemented group. Data are mean $\pm$ SEM values of eight rats.

^aSignificantly different compared with the value for SC rats by the independent samples t test at $P < .01$.

^bSignificantly different compared with the value for EC rats by the independent samples t test at $P < .05$.

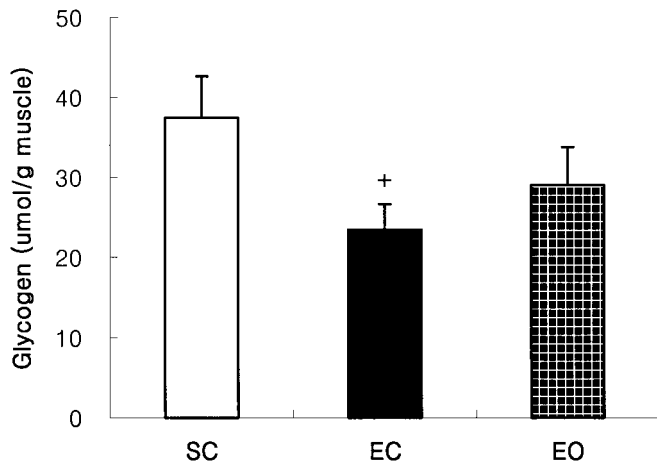


FIG. 2. Muscle glycogen concentration of exhausted rats following the running performance test: SC, sedentary control group; EC, exercise-trained control group; EO, exercise-trained, octacosanol-supplemented group. Data are mean $\pm$ SEM values of eight rats. ⁺Significantly different compared with the value for SC rats by the independent samples *t* test at $P < .05$.

immediately following the running performance test. A significant decrease (30%) in plasma glucose concentration was observed in the exercise-trained rats compared with the SC animals ($P < .01$). Although the exercise-trained rats ran 1.8 times longer until exhausted than did the sedentary rats, their plasma levels of ammonia, lactate, and inorganic phosphorus were not significantly different from the SC group. Significant elevations in plasma ammonia and lactate concentrations were found in the EO rats, which ran 46% longer until exhausted, compared with the EC rats ($P < .05$), while the octacosanol-induced decrease in plasma glucose concentration of exercise-trained rats following the running performance test was not statistically significant ($P > .05$). Plasma inorganic phosphorus concentration was not affected by either the exercise training or the octacosanol supplementation.

Gastronecnius muscle glycogen concentration under the exhausted state was lower in the exercise-trained rats than

the value for the SC group ($P < .05$), but was not significantly different between the EC and EO rats, despite the latter group running 45% longer until exhausted (Fig. 2).

Table 6 shows the activities of selected enzymes measured in the plasma and skeletal muscle of exhausted rats. Plasma CK activity was significantly affected by both the exercise training and diet: Rats in the EC group had significantly higher CK activity (198 ± 19.55 IU/L) than did the SC rats (128 ± 7.85 IU/L) ($P < .01$), and EO rats had significantly higher plasma CK activity (285 ± 20.72 IU/L) compared with the EC animals ($P < .01$). Four weeks of exercise training without dietary supplementation resulted in significantly higher CS activity in the soleus muscle of rats (15%, $P < .01$) compared with the value for the SC rats. Feeding the octacosanol-supplemented diet in trained rats resulted in higher muscle CS activity than observed in the trained control animals (16% increase, $P < .01$). The LDH activity in the gastrocnemius muscle of rats exhibited a similar tendency as shown for the activities of CK and CS, in that it tended to be increased by the exercise training (25.6 ± 1.94 vs. 19.5 ± 1.30 $\mu\text{mol mg of muscle}^{-1} \text{ min}^{-1}$) and further increased by the dietary supplement (32.4 ± 1.94 vs. 25.6 ± 1.94 $\mu\text{mol mg of muscle}^{-1} \text{ min}^{-1}$), although these training and dietary effects were not statistically significant ($P > .05$).

DISCUSSION

Peripheral factors that contribute to exercise-induced fatigue such as glycogen depletion and lactate accumulation have been extensively studied and are well described.¹⁷ Lactate is produced during glycolysis, and under normal circumstances it is further degraded via the tricarboxylic acid cycle for the production of ATP by oxidative phosphorylation. There is also growing evidence of a significant central (*i.e.*, a central nervous system) component, such as plasma ammonia level, to exercise-induced fatigue.^{18,19} The immediate source of ammonia from muscle appears to be a result of the deamination of AMP and is more apparent in fast-twitch than in slow-twitch fibers. In high-intensity exercise, the increase in ADP concentration activates the enzyme AMP deaminase, a key enzyme in the purine nucleotide cy-

TABLE 6. ACTIVITIES OF SELECTED ENZYMES IN SKELETAL MUSCLE AND PLASMA OF EXHAUSTED RATS FOLLOWING THE RUNNING PERFORMANCE TEST

	Group		
	SC	EC	EO
CK (IU/L of plasma)	128 $\pm$ 7.85	198 $\pm$ 19.6 ^a	285 $\pm$ 20.7 ^b
CS (nmol mg of soleus muscle ⁻¹ min ⁻¹)	46.9 $\pm$ 2.00	54.1 $\pm$ 1.32 ^c	62.6 $\pm$ 2.48 ^b
LDH ($\mu\text{mol mg of gastrocnemius muscle}^{-1} \text{ min}^{-1}$)	19.5 $\pm$ 1.30	25.6 $\pm$ 2.73	32.4 $\pm$ 1.94

SC, sedentary control group; EC, exercise-trained control group; EO, exercise-trained and octacosanol-supplemented group. Data are mean $\pm$ SEM values of eight rats.

^aSignificantly different compared with the value for SC rats by the independent samples *t* test at $P < .05$, ^a $P < .01$.

^bSignificantly different compared with the value for EC rats by the independent samples *t* test at $P < .01$.

cle, leading to an increased production of NH_4^+ .²⁰ Increases in blood ammonia levels have been reported in rats after swimming and in humans after arm work, maximal cycle ergometry, and treadmill exercise.^{20–22}

The effect of the training protocol used in this study is confirmed by the increased time to reach exhaustion observed for the trained rats without dietary supplementation. The training protocol in our study also increased the gastrocnemius muscle (fast-twitch fiber) weight and CS activity in the soleus muscle (slow-twitch fiber) of rats, both of which are known to be elevated by aerobic exercise training.^{23–25} In this study, the exercise-trained rats that were not supplemented appeared to accumulate ammonia and lactate in their blood at a slower rate than did SC animals, so they could run longer until exhausted. These data are in agreement with human studies showing that trained subjects generally experience a lesser increase in plasma ammonia during either prolonged or intense exercise.²⁶ The major metabolic consequences of adaptation to endurance exercise training are a slower utilization of muscle glycogen and blood glucose, a greater reliance on fat oxidation, and less lactate production in the muscle during exercise of a given intensity.²⁷ The presence in the blood of enzymes that are normally localized in muscle fibers is taken as evidence of disruption or increased permeability of the muscle cell membranes. In this respect, increased plasma CK is considered one of the best indicators of muscle damage from eccentric contraction.²⁸ Exercise training is known to prevent the rise of plasma CK during a single bout of exercise under a given protocol.^{29,30} The reason our trained rats exhibited higher plasma CK activity appears to be attributable to the increased running time to exhaustion compared with the SC rats.

We investigated the effect of octacosanol supplementation on the time to reach exhaustion of exercise-trained rats subjected to a single bout of high-intensity exercise. We chose a 0.75% octacosanol supplementation level based on earlier reports by other investigators^{3,31} that octacosanol (or polycosanol) was effective in preventing myocardial necrosis and modulating lipid metabolism in rats at 0.1–1% levels. Octacosanol supplementation produced an increase in the time to exhaustion of exercise-trained rats. Elevations in plasma lactate and ammonia concentrations and CK activity following the running performance test appear to be related to the increase in endurance. Although EO rats ran significantly longer until exhausted, their plasma glucose and gastrocnemius muscle glycogen concentrations under the exhausted state were not significantly different from those for EC rats. Octacosanol supplementation significantly elevated CK and CS activities in the muscle of exercise-trained rats. These results can be summarized as follows: Octacosanol supplementation *per se* increased the endurance exercise performance of trained rats without down-regulating the production of lactate and ammonia. Therefore, octacosanol exerts its ergogenic activity through other biochemical mechanisms involving increased activity of CS

and enhanced glycogen storage (and/or saving of glycogen utilization) in trained muscle.

ACKNOWLEDGMENTS

This study was supported by the Technology Development Program for Agriculture and Forestry, Ministry Agriculture and Forestry, Republic of Korea.

REFERENCES

1. Beltz SD, Doering PL: Efficacy of nutritional supplements used by athletes. *Clin Pharm* 1993;12:900–908.
2. Saint-John M, McNaughton L: Octacosanol ingestion and its effects on metabolic responses to submaximal cycle ergometry, reaction time and chest and grip strength. *Int Clin Nutr Rev* 1986;6:81–87.
3. Kato S, Karino K, Hasegawa S, *et al.*: Octacosanol affects lipid metabolism in rats fed on a high-fat diet. *Br J Nutr* 1995;73:433–441.
4. Canetti M, Moreira M, Mas R, *et al.*: A two-year study on the efficacy and tolerability of policosanol in patients with type II hyperlipoproteinaemia. *Int J Clin Pharmacol Res* 1995;15:159–165.
5. Menendez R, Arruzazabala L, Mas R, *et al.*: Cholesterol-lowering effect of policosanol on rabbits with hypercholesterolaemia induced by a wheat starch-casein diet. *Br J Nutr* 1997;77:923–932.
6. Mas R, Castano G, Illnait J, *et al.*: Effects of policosanol in patients with type II hyperlipoproteinaemia and additional coronary risk factors. *Clin Pharmacol Ther* 1999;65:439–447.
7. Castano G, Mas R, Fernandez J, *et al.*: Effects of policosanol on postmenopausal women with type II hyperlipoproteinaemia. *Gynecol Endocrinol* 2000;14:187–195.
8. McCarty MF: Policosanol safely down-regulates HMG-Co A reductase-potential as a component of the Esselstyn regimen. *Med Hypothesis* 2002;59:268–279.
9. Castano G, Mas R, Fernandez JC, Fernandez L, Illnait J, Lopez E: Effects of policosanol on older patients with hypertension and type II hyperlipoproteinaemia. *Drugs R D* 2002;3:159–172.
10. Cureton TK: Forty-two physiological training program utilizing 894 humans. In: *The Physiological Effects of Wheat Germ Oil on Humans. In Exercise*, Charles C Thomas, Springfield, IL, 1972, pp. 525–715.
11. Levin E: Effects of octacosanol on chick comb growth. *Proc Soc Exp Biol Med* 1963;112:331–334.
12. Passwater R: Octacosanol. Its name really spells increased energy, stamina and vigor. *Health Q* 1982;7:14–15.
13. Institute of Laboratory Animal Resources, National Research Council: *Guide for the Care and Use of Laboratory Animals*, National Academy Press, Washington, DC, 1996.
14. Srere PA: Citrate synthase. *Methods Enzymol* 1969;13:3–11.
15. Pesce A, Fondy TP, Stolzenbach F, Castillo F, Kaplan NO: Comparative enzymology of lactate dehydrogenases. 3. Properties of the H4 and M4 enzymes from a number of vertebrates. *J Biol Chem* 1967;242:2151–2167.
16. Chun Y, Yin ZD: Glycogen assay for diagnosis of female genital *Chlamydia trachomatis* infection. *J Clin Microbiol* 1998;36:1081–1082.
17. Gollnick PD: Peripheral factors as limitations to exercise capacity. *Can J Appl Sports Sci* 1982;7:14–21.
18. Davis JM, Bailey SP: Possible mechanisms of central nervous system fatigue during exercise. *Med Sci Sports Exerc* 1997;29:45–57.

19. Mutch BJC, Banister EW: Ammonia metabolism in exercise and fatigue: a review. *Med Sci Sports Exerc* 1983;15:41–50.
20. Katz A, Broberg S, Sahlin K, Wahren J: Muscle ammonia and amino acid metabolism during dynamic exercise in man. *Clin Physiol* 1986;6:365–379.
21. Denis C, Linossier MT, Dormois D, Cottier-Perrin M, Geyssant A, Lacour JR: Effects of endurance training on hyperammoniaemia during a 45-min constant exercise intensity. *Eur J Appl Physiol* 1989;59:268–272.
22. Graham TE, Turcotte LP, Kiens B, Richter EA: Training and muscle ammonia and amino acid metabolism in humans during prolonged exercise. *J Appl Physiol* 1995;78:725–735.
23. Meneguello MO, Mendonca JR, Lancha AH Jr, Costa Rosa FBP: Effect of arginine, ornithine and citrulline supplementation upon performance and metabolism of trained rats. *Cell Biochem Funct* 2003;21:85–91.
24. Jansson E, Kaijser L: Substrate utilization and enzymes in skeletal muscle of extremely endurance-trained men. *J Appl Physiol* 1987;62:999–1005.
25. Siu PM, Donley DA, Bryner RW, Always SE: Citrate synthase expression and enzyme activity after endurance training in cardiac and skeletal muscles. *J Appl Physiol* 2003;94:555–560.
26. Graham TE, Turcotte LP, Kiens B, Richter EA: Effect of endurance training on ammonia and amino acid metabolism in humans. *Med Sci Sports Exerc* 1997;29:646–653.
27. Crowley MA, Willis WT, Matt KS, Donovan CM: A reduced lactate mass explains much of the glycogen sparing associated with training. *J Appl Physiol* 1996;81:362–367.
28. Armstrong RB, Ogilvie RW, Schwane JA: Eccentric exercise-induced injury to rat skeletal muscle. *J Appl Physiol* 1983;54:80–93.
29. Schwane JA, Armstrong RB: Effect of training on skeletal muscle injury from downhill running in rat. *J Appl Physiol* 1983;55:969–975.
30. Hyatt JP, Clarkson PM: Creatine kinase release and clearance using MM variants following repeated bouts of eccentric exercise. *Med Sci Sports Exerc* 1998;30:1059–1065.
31. Herrera MN, Magraner J, Mas R: Effect of policosanol on isoprenaline-induced myocardial necrosis in rats. *J Pharm Pharmacol* 1994;46:282–285.